

Lab 17: Exam 03 Review — Modules 12–15

BIOL-1

Name: _____ Date: _____

Purpose

This lab is **practice** for the third unit: **evolution in populations, speciation, and ecology**. It pulls together ideas from your Keys to Success and class notes. It is **not** a copy of the official exam; wording and topics are reordered and rephrased so you practice **concepts**, not one memorized set of sentences.

How to use this lab

1. Do the **written** items **closed-book** first, then reopen notes only where you were stuck. A short **check** (A–D) list at the end of each module is optional—use it to verify you still recognize exam-style phrasing.
2. For **short answers**, write in full sentences (or labeled bullet lists where that matches your instructor's expectations). **Definitions** should use **your** words, not a textbook sentence—then compare to the key terms list.
3. Ask your instructor for the **official exam** instructions, timing, and point breakdown—those rules override anything informal here.

Module 12 — Darwin and evolution

Focus ideas

- **Descent with modification** (change over time from shared ancestors).
- **Natural selection:** heritable **variation**, **overproduction** of offspring, **differential reproductive success** (fitness = relative reproductive success).
- **Individuals** do not evolve; **populations** change in allele frequency over generations.
- **Evidence** from fossils, biogeography, comparative anatomy (homology), molecular comparisons.

Written practice (free response)

1. In **one** clear sentence, define **evolutionary fitness** the way a biologist uses the term in this unit (not “strongest” or “survives longest” unless you explain why that could still matter to fitness).

2. In **2–3** sentences, explain “**descent with modification**” so a friend in another class would understand: ancestry, **change over time**, and why **not** all descendants look identical.

3. **Scenario:** A tall plant and a short plant in the **same** population both survive a drought **equally well**, but the tall one produces more seeds. In **2–3** sentences, say what is acting as the **selective pressure**, what **variation** the selection is sorting, and what **differential reproductive success** means here.

4. Name **one** line of **evidence** for evolution (fossil, comparative anatomy, biogeography, or molecular) and state **what pattern** that evidence shows—**two** sentences max.

5. In **2–3** sentences, explain why “**populations** evolve, not individuals” using **allele frequencies** (or gene-pool change), not the word *fitness* alone.

6. **Optional contrast: Acclimation** (one body over time) vs **evolution** (across generations). **One** sentence each.

Quick check — write the letter (A–D) *optional*

A. In biology, **evolutionary fitness** is best defined as:

- A) Physical strength relative to other species
- B) Survival time without food
- C) Relative reproductive success (passing alleles to the next generation)
- D) Ability to acclimate to cold weather in one lifetime

B. Which is **strong** evidence of **common descent**?

- A) Two species on different continents, so they must be unrelated
- B) The same **homologous** bone layout in forelimbs of a whale, a bat, and a human, with different functions
- C) Unrelated lineages that look alike because of convergent selection
- D) A single individual that lived 200 years and changed its traits

Module 13 — How populations evolve (microevolution)

Focus ideas

- **Microevolution** = **change in allele frequencies** in a population over time.
- **Five mechanisms:** mutation, **gene flow**, **genetic drift**, non-random mating (e.g. sexual selection), **natural selection**.
- **Drift** matters most in **small** populations; **bottleneck** vs **founder** effects.
- **Selection modes** on a continuous trait: **directional**, **stabilizing**, **disruptive**.
- **Hardy–Weinberg** = mathematical baseline of **no** evolution; real populations usually violate it.

Written practice (free response)

1. Mechanisms list: List the **five** processes that can change **allele frequencies** in a population. Then circle (in your mind) the **two** that are **not** “selection” in the usual sense; name those two in one sentence and say how they differ from each other in **one** more sentence.

2. Drift vs selection: A hurricane **randomly** kills 90% of a moth population; survivors repopulate. In **2–3** sentences, say whether this is best viewed as **selection** or **drift** and name the specific **drift** label if it applies. Then give a **separate** two-sentence example where **differential survival of genotypes** is **selection**, not random sampling.

3. Founder vs bottleneck: A few lizards **found** a new island population; a storm **bottlenecks** a large mainland population. In **2–3** sentences, contrast what each event does to **allele variety** in the long run (same? different? why?).

4. Selection shapes: For **each** of **directional**, **stabilizing**, and **disruptive** selection, give **one** short phrase: what the environment favors (e.g. “favors the middle of a birth-weight range”).

5. Hardy–Weinberg: In **2–3** sentences, say what the **equilibrium** is **used for** in this course (a word like **null** or **baseline** helps) and name **two** real-world violations that would break it (not a full list—**two** is enough).

6. Two terms you confuse (your words):

Quick check — write the letter (A–D) optional

A. Which option is a **mechanism of microevolution**?

A) A sunburn that is not passed to children

- B) **Gene flow** when individuals enter the population and mate
- C) A population count going up because of immigration without allele change
- D) A species inventing a new organ when needed

B. A few lizards are accidentally transported to a new island; the new population only reflects the alleles the founders carried. This is:

- A) The **founder** effect (genetic drift)
 - B) The bottleneck effect in the source population on the mainland
 - C) Convergent evolution
 - D) Interspecific competition
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Module 14 — Macroevolution and speciation

Focus ideas

- **Biological species concept** (interbreed in nature → viable fertile offspring; **reproductive isolation** from other such groups).
- **Prezygotic** barriers (before zygote): habitat, **temporal**, behavioral, mechanical, gametic, etc.
- **Postzygotic** barriers (after zygote): hybrid viability, hybrid fertility, hybrid breakdown.
- **Allopatric** speciation: geographic split then divergence. **Sympatric** speciation: same place (e.g. **polyploidy** in plants).

Written practice (free response)

1. Prezygotic vs postzygotic: For **(a)** different spawning months, **(b)** hybrid offspring sterile, and **(c)** different courtship songs, label each as **prezygotic** or **postzygotic** and give a **one-word** subtype where you can (temporal, behavioral, etc.).

2. Horse × donkey → mule: In **2–3** sentences, explain which kind of **reproductive barrier** the **sterile mule** represents and **why** that still **blocks** gene flow between the parent species over time.

3. Allopatric pattern: A mountain range splits one species into two populations that no longer **exchange** genes. In **2–3** sentences, describe the **allopatric** story (geography + time + divergence) and one **kind** of difference that might accumulate in each lineage.

4. Sympatric in plants: Explain **sympatric** speciation by **polyploidy** in **2–3** sentences: same place, new chromosome set, and why isolation can be **sudden** compared with slow geographic split.

5. Macroevolution definition: In **2–3** sentences, explain why “macroevolution” in this course is **not** a separate *kind* of DNA magic from “microevolution” but **larger-scale** outcomes built from the same population-level processes.

6. Reproductive isolation and divergence: In **2–3** sentences, explain why, once two lineages **do not** swap genes, they can **accumulate different** adaptations in different environments (use **allele** or **trait** change language).

Quick check — write the letter (A–D) *optional*

A. A horse and a donkey can produce a mule, but the mule is typically sterile. This outcome is:

- A) A prezygotic gametic failure
- B) A **postzygotic** barrier to gene flow (reduced hybrid fertility)
- C) Stabilizing selection in horses
- D) The founder effect

B. A mountain range forms and cuts a **single** species into two populations that no longer exchange genes. Over time they diverge until they do not interbreed. This pattern fits:

- A) **Allopatric** speciation
- B) Only sympatric speciation with polyploidy
- C) Macroevolution without microevolution
- D) A species concept based only on appearance

Module 15 — Population and systems ecology

Focus ideas

- **Exponential (J)** vs **logistic (S)** growth; **carrying capacity (K)**.
- **Density-dependent** (competition, disease, predation often intensify with crowding) vs **density-independent** (e.g. many fires, some storms) regulation.

- **Trophic levels** and ~10% **energy** transfer; **energy flows, nutrients** cycle.
- **Producers, consumers, decomposers**; food chains limited in length partly because of energy loss as **heat** and work.

Written practice (free response)

1. J vs S and K: In 3–4 sentences, describe how a **logistic** curve **differs** from unrestrained **exponential** growth, and what **K** means for a real population (you may use a short example: elk, plants, or bacteria).

2. Density (two examples): Give **one** example of a **density-dependent** limit and **one** of a **density-independent** disturbance, each in a **full sentence** with **why** the label fits.

3. Energy vs nutrients: Contrast **energy** flow and **nutrient** cycling in an ecosystem in 3–4 sentences, including decomposers and **why** sunlight is always the **new** energy input.

4. Ten percent in words: With ~10% transfer per trophic level, explain in 2–3 sentences why supporting **the same** number of people with **tuna** (long chain) generally requires **more** primary production than eating **closer to plants**—clear logic, no need for exact numbers if your reasoning is sound.

5. Connect K and energy (one sentence, your words): Link **carrying capacity** to **trophic** levels—why a big base of **producers** supports **few** top consumers.

Quick check — write the letter (A–D) *optional*

A. A disease spreads more easily when hosts are very crowded. That limiting factor is best classified as:

- A) Density-**independent**
- B) Density-**dependent**
- C) Only abiotic (never biological)
- D) Not part of population ecology

B. In ecosystems, **chemical elements** (e.g. nitrogen, carbon in nutrients) are **cycled**, while **energy** is:

- A) Cycled the same way as nitrogen
- B) **Lost as heat** along the way; not recycled to be photosynthesized again from dead matter in the same way sunlight is new input
- C) Only stored in apex predators
- D) Only found in the atmosphere, never in plants

| Word bank sprint

Use each term once. Practice meaning in context, not one memorized sentence shape.

Word bank: *allele frequencies, allopatric, biological species, density-dependent, descent with modification, decomposers, directional selection, homologous, prezygotic, stabilizing selection*

1. In formal terms, **microevolution** means a change in _____ from one generation to the next in a population.
2. In comparative anatomy, forelimb bones in different tetrapods that share structure from a common ancestor are called _____ structures (even if functions differ).
3. A population's birth-weight curve can favor the middle; this mode of natural selection is _____ selection.
4. A barrier that works **before** a zygote forms—such as different breeding seasons—is a _____ barrier to reproduction.
5. A river splits a range; populations diverge in isolation—speciation that begins with **geography** is often _____ speciation.
6. Darwin's phrase for evolution: species share ancestry and accrue change over time—_____ with modification.

7. A limiting factor that hits harder as a population **crowds** (e.g. more transmission of a parasite) is _____ regulation in many textbook examples.
8. A group of populations that interbreed in nature, producing viable fertile offspring, and is reproductively isolated from other such groups, fits the _____ species concept (as presented in this unit).
9. Fungi and bacteria that recycle dead organic matter and return elements to the soil for producers are _____.
10. When a population's average trait value shifts toward **one** extreme of the range (e.g. darker moths on sooty trees), the pattern is _____ selection.

Short-answer rehearsal

Draft at least three in your own words. On your exam, you may answer a subset of longer prompts; check your real instructions.

1. (Module 12) Contrast how **individuals** can change in a lifetime with how **evolution** is defined in this course. Name **one** type of **evidence** for evolution in one sentence.

2. (Module 13) Compare **genetic drift** to **natural selection** (what drives change in each?). Give **one** example of **bottleneck** or **founder** effect in plain words.

3. (Module 14) Classify these as **prezygotic** or **postzygotic**: (a) different courtship song; (b) hybrid is sterile. Then state one difference between **allopatric** and **sympatric** speciation in one or two sentences.

4. (Module 15) Draw the distinction between **density-dependent** and **density-independent** limits, then use the **~10% rule** to explain why long food chains support **fewer**

high-level consumers than a base of the same size could support if humans ate **primary producers** directly. (Equations optional; clear logic is enough.)

5. (Synthesis, Module 12–13) A small island population of birds loses genetic diversity after a storm. In **2–3** sentences, explain **one** way this could be **drift** and **one** way a biologist would **not** call the same event “selection.”

6. (Synthesis, Module 14–15) A **physical barrier** (new canyon) splits a fish population, and on one side, **prey** are scarce. In **2–3** sentences, tie **allopatric** divergence to how **K** or **trophic** pressure could **differ** between the two sides over time (conceptual, not a math problem).

Self-check before the exam

I can...	✓
State natural selection as variation + overproduction + unequal contribution to the next generation	[]
Give one type of evidence (fossil, anatomy, biogeography, molecular) and what it supports	[]
Name the five microevolution mechanisms; contrast drift vs gene flow	[]
Describe directional / stabilizing / disruptive selection; explain bottleneck vs founder	[]
Classify prezygotic vs postzygotic; allopatric vs sympatric; biological species idea	[]
	[]

I can...	✓
Contrast J vs S growth; K; density-dependent vs independent; energy flow vs nutrient cycling ; 10% rule	
Two terms I still mix up: 	
One action before the exam (be specific):	

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